

Dietary fat and meat intakes and risk of reflux esophagitis, Barrett's esophagus and esophageal adenocarcinoma

The aim of this study was to investigate whether dietary fat and meat intakes are associated with reflux esophagitis (RE), Barrett's esophagus (BE) and esophageal adenocarcinoma (EAC). In this all-Ireland case-control study, dietary intake data was collected using a food frequency questionnaire in 219 RE patients, 220 BE patients, 224 EAC patients, and 256 frequency-matched controls between 2002 and 2005. Unconditional multiple logistic regression analysis was used to examine the association between dietary variables and disease risk using quartiles of intake, to attain odds ratios (OR) and 95% confidence intervals (95%CI), while adjusting for potential confounders. Patients in the highest quartile of total fat intake had a higher risk of RE (OR=3.54; 95%CI=1.32–9.46) and EAC (OR=5.44; 95%CI=2.08–14.27). A higher risk of RE and EAC was also reported for patients in the highest quartile of saturated fat intake (OR=2.79; 95%CI=1.11–7.04; OR=2.41; 95%CI=1.14–5.08, respectively) and monounsaturated fat intake (OR=2.63; 95%CI=1.01–6.86; OR=5.35; 95%CI=2.14–13.34, respectively). Patients in the highest quartile of fresh red meat intake had a higher risk of EAC (OR=3.15; 95%CI=1.38–7.20). Patients in the highest category of processed meat intake had a higher risk of RE (OR=4.67; 95%CI=1.71–12.74). No consistent associations were seen for BE with either fat or meat intakes. Further studies, investigating the association between dietary fat and food sources of fat are needed to confirm these results.